

Sketch-in-Scene: Coherent 3D Scene Asset Generation with Sketches

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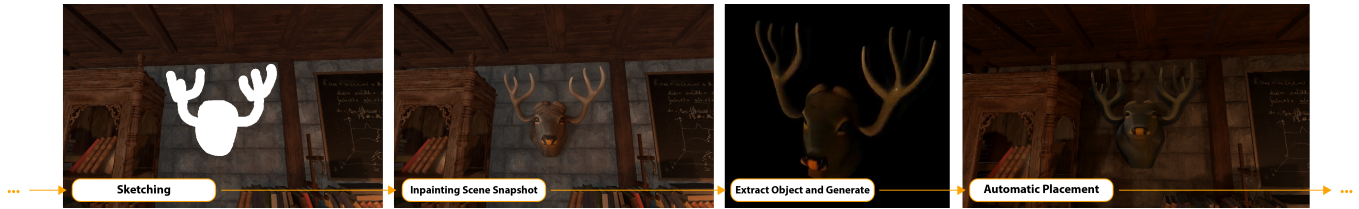


Figure 1: Sketch-in-scene is a system that enables quick generation of 3D assets with scene context. The user first sketches in the scene; then, the snapshot of the screen will be inpainted. The object of interest will be extracted and converted into a 3D object. Last, the asset will be automatically placed back in the scene, which facilitates quick iterative authoring of the scene.

Abstract

This paper introduces an innovative sketch- and environment-based 3D asset generation system to generate assets that are coherent with the surroundings. The key idea is to not directly utilize the image or text as conditions, but rather to use the inpainted part in a scene’s snapshot as the condition for 3D generation. The system utilizes in-3D sketches and quickly generates the scene-coherent 3D assets based on the users’ text prompt. Our solution has a frontend for the user to navigate in 3D and perform sketching, and a backend that generates an inpainted image for the scene, extracts the object-of-interest and generates the 3D asset only with the inpainted part of the image. The generated asset can then be automatically placed back to the 3D scene based on heuristics. Our work can not only generate coherent assets, but also demonstrate a simple and efficient workflow to create coherent 3D assets in the virtual environment.

CCS Concepts

• Computing methodologies → Graphics systems and interfaces.

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1 Introduction

The conventional pipeline of creating a 3D scene via 3D modeling and scene editing remains labor-intensive and time-consuming,

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often hindering both the efficiency and accessibility of 3D content production. These challenges may significantly diminish user motivation, particularly for those seeking to rapidly prototype or iterate on 3D scenes. As such, 3D generation works, like TRELLIS [Xiang et al. 2025], have captured interest broadly from academia and industry, and some studies have further investigated generating 3D scenes through text or image instruction [Bokhovkin et al. 2025; Öcal et al. 2024; Yan et al. 2024]. There is, however, a need to improve fine-grained control to improve human agency.

It can be difficult to quickly populate coherent 3D assets for a scene. Some studies have employed a method of combining sketching and 3D scene generation. For example, previous work has investigated generating a scene roughly based on sketch input [Xu et al. 2024] and improving the quality of generation with sketches [Xiang et al. 2025], but these works still lack the fine-grained control for the scene. Of course, on the other hand, there are also works that can generate high-quality 3D models [Xiang et al. 2025] and further conditioned with sketching [Liu et al. 2024] and proxies [Dong et al. 2024]. Users can generate and place the object one-by-one to a scene, but that could be cumbersome and more importantly, the generated objects may not be coherent with each other and the surrounding scene. We can see that there is a gap between scene generation and 3D object generation currently — a lack of investigation on how to generate 3D objects that are contextualized by the surroundings.

We propose *Sketch-in-Scene* (Figure 1), a system that generates 3D objects based on the scene context. The user can sketch in the virtual environment and use a text prompt to describe the object, and the 3D generation model will generate the object that is coherent with the surroundings

2 Sketch-in-Scene System

As shown in Figure 2, the proposed scene-aware sketch-based 3D asset generation system mainly consists of three parts: (1) the sketching frontend that enables the user to draw with a 3D brush and

navigate the scene; (2) an image generation model that takes in the masked scene snapshot and prompt to generate an image condition for the object; and, (3) a 3D object generation model that is conditioned only with the relevant inpainted part of the object.

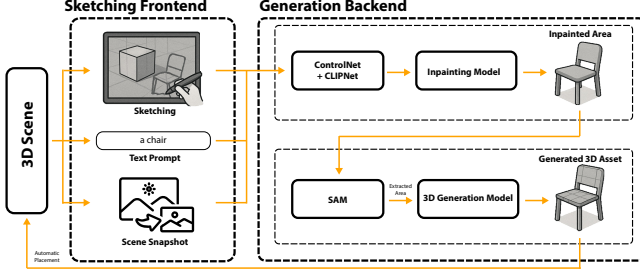


Figure 2: System overview.

The interactive **sketching frontend** provides several functions, such as navigation, object manipulation (for poses of objects), and of course, sketching with a 3D brush. To generate a new asset, the user can first use a 3D brush and draw a sketch in the virtual scene; then, the user needs to provide a brief text prompt to describe the asset. When the sketch is completed and confirmed, the 3D sketch and camera pose information are saved, and the text prompt, snapshot of the scene (snapshot from the user’s perspective) and sketch (also from the user’s perspective) are fed to the backend models. After the backend models have generated the coherent 3D asset, the frontend will automatically place the object based on heuristics. Specifically, the automatic placement procedure will rotate the generated model such that its normal vector of the largest surface aligns with the largest surface normal vector of 3D sketch’s bounding box. Similarly, we can align the asset’s normal vector of the smallest surface with the 3D sketch’s one. After alignment, the model will be scaled and translated to achieve automatic placement in a scene.

To create coherence with the surroundings, the idea is to use the snapshot of the scene as a base and use the sketch data (including prompts) for **inpainting the masked area** (based on alpha values in the sketches) to create an image of a new object. The multimodal diffusion model [Batifol et al. 2025; Liu et al. 2025] that combines image generation, ControlNet and Stable Diffusion is used to generate for the inpainted area, which includes the 2D view of the asset. Meanwhile, we also used the Contrastive Language–Image Pre-training (CLIP) model [Radford et al. 2021] to analyze and classify the style of background snapshots. This model guides the generation of 3D assets that simultaneously meet environmental coherence and style coherence.

Then, we move to utilizing the 2D inpainted images for **generating coherent 3D assets**. It is critical to extract the object-of-interest from the inpainted area. We use the Segment Anything model [Kirillov et al. 2023] to achieve high-quality segmentation. By randomly sampling several pixel points in the central area ($< 1/2$ radius) of the mask, we use a majority vote method to select the segmentation results that exceeded half of the total as the target object region, thereby extracting the best representative object image. The extracted object-of-interest (an image) is then fed to our 3D

object generation model of choice, TRELLIS, for image-conditioned 3D generation. Recall that the image of the 3D object is extracted from the inpainted area of the scene’s snapshot; thus, the asset generated with our model has better visual coherence than simply using TRELLIS with either text and image input (Figure 3). This 3D asset will then be passed back to the front-end for placing into the scene.

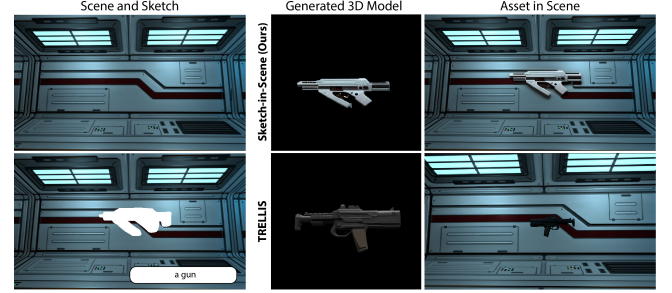


Figure 3: Sketch-in-Scene is able to generate result that is more coherent with the surroundings. Top row is our proposed model while the bottom row is TRELLIS only.

3 Result and Conclusion

Compared with other methods, the novelty of this system lies in not directly using the text and image condition for 3D generation, but to use the extracted part of the inpainted snapshot of the scene to inject scene-context for coherent asset generation. The result is that the generated 3D asset is visually more coherent with the surroundings (Figure 4). The backend of the system has been tested on an RTX 4090 computer with 24GB VRAM. The average processing time to generate is about 20s.

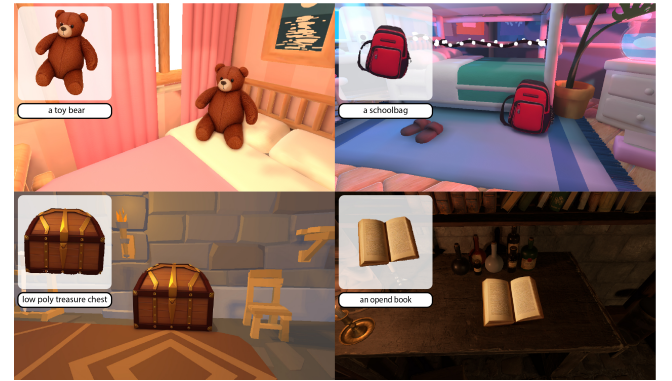


Figure 4: Our model can be used in various scenarios. Top-left is the extracted inpainted area and the text prompt.

To conclude, we believe our work has showcased a novel and quick workflow to coherently generate 3D asset for a virtual scene. We hope that it can inspire future tools and work. In addition, we believe our work highlights a research gap — 3D object generation with scene context — that is worth investigating.

References

- Stephen Batifol, Andreas Blattmann, Frederic Boesel, Saksham Consul, Cyril Diagne, Tim Dockhorn, Jack English, Zion English, Patrick Esser, Sumith Kulal, et al. 2025. FLUX. 1 Kontext: Flow Matching for In-Context Image Generation and Editing in Latent Space. *arXiv e-prints* (2025), arXiv–2506.
- Aleksey Bokhovkin, Quan Meng, Shubham Tulsiani, and Angela Dai. 2025. Scenefactor: Factored latent 3d diffusion for controllable 3d scene generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 628–639.
- Wenqi Dong, Bangbang Yang, Lin Ma, Xiao Liu, Liyuan Cui, Hujun Bao, Yuewen Ma, and Zhaopeng Cui. 2024. Coin3d: Controllable and interactive 3d assets generation with proxy-guided conditioning. In *ACM SIGGRAPH 2024 Conference Papers*. 1–10.
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4015–4026.
- Feng-Lin Liu, Hongbo Fu, Yu-Kun Lai, and Lin Gao. 2024. Sketchdream: Sketch-based text-to-3d generation and editing. *ACM Transactions on Graphics (TOG)* 43, 4 (2024), 1–13.
- Zichen Liu, Yue Yu, Hao Ouyang, Qiuyu Wang, Ka Leong Cheng, Wen Wang, Zhiheng Liu, Qifeng Chen, and Yujun Shen. 2025. Magicquill: An intelligent interactive image editing system. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 13072–13082.
- Başak Melis Öcal, Maxim Tatarchenko, Sezer Karaoğlu, and Theo Gevers. 2024. Sceneteller: Language-to-3d scene generation. In *European Conference on Computer Vision*. Springer, 362–378.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PmLR, 8748–8763.
- Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. 2025. Structured 3d latents for scalable and versatile 3d generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 21469–21480.
- Yongzhi Xu, Yonhon Ng, Yifu Wang, Inkyu Sa, Yunfei Duan, Yang Li, Pan Ji, and Hongdong Li. 2024. Sketch2Scene: Automatic Generation of Interactive 3D Game Scenes from User’s Casual Sketches. *arXiv preprint arXiv:2408.04567* (2024).
- Ziyang Yan, Lei Li, Yihua Shao, Siyu Chen, Zongkai Wu, Jenq-Neng Hwang, Hao Zhao, and Fabio Remondino. 2024. 3dsceneeditor: Controllable 3d scene editing with gaussian splatting. *arXiv preprint arXiv:2412.01583* (2024).